

Manthan Bhala

Machine Learning Engineer & Software Developer

Gurugram, Haryana, India
manthanbhala@gmail.com
linkedin.com/in/manthan-bhala-a11760136
github.com/ManthanBhala

EXPERIENCE

Senior Machine Learning Engineer — Makro PRO

Oct 2023 – Present

Remote

- Developed e-commerce recommendation models that increased total basket additions by 52%.
- Built personalisation models by distilling LLMs into LightGBM for production-scale inference.
- Built a vector search system for e-commerce with p99 latency under 100ms using vLLM.
- Improved search results through autocategorization, autocorrect, query expansion, and query rewriting models.
- Built a chatbot POC with RAG pipeline and function calling, optimized with quantization and vLLM deployment.

Software Developer 2 — Frinks AI

Mar 2023 – Jul 2023

Bengaluru, Karnataka, India

- Developed and optimized ML inference pipelines for real-time defect detection and counting on industrial production lines.
- Built computer vision and backend services for a quality inspection platform for dimensioning, surface defect, and crack detection.

Software Developer — Jaguar Land Rover

Sep 2020 – Feb 2023

Bengaluru, Karnataka, India

- Developed and maintained ML pipelines and backend services for Vehicle Algorithmic Self Diagnostics Tool.
- Worked on Advanced Driver Assistance Systems (ADAS) for autonomous and assisted driving features.

SKILLS

ML Engineering

Python · PyTorch · TensorFlow · Hugging Face

Applied AI

Information Retrieval · Learning to Rank · Recommendation Systems · LLMs · PEFT

Data & Infrastructure

SQL · PySpark · Docker · Kubernetes · CI/CD · Redis · Vector Search

EDUCATION

Bachelor of Technology — Indian Institute of Technology (IIT), Bombay 2016 – 2020 | CGPA: 8.48/10

INTELLECTUAL PROPERTY

Patent 2501007631 — A Hybrid Lexical-Semantic Retrieval Process for E-Commerce Search

Patent 2501004476 — A Hybrid LLM Process for Multi-Objective Reranking in E-Commerce

Patent 2501004419 — A Multilingual Phonetic BK-Tree Based Spelling Correction Process in E-Commerce